

Upstream Production Optimization & Predictive BOE Analytics Report

Project by Fatoye Yetunde

Executive Summary

This project was undertaken to evaluate the operational and financial performance of upstream oil and gas assets by analyzing production output, reservoir health, operating costs, revenue generation, and profitability. The analysis integrates BOE performance indicators with financial metrics to provide a comprehensive view of asset value creation and support data-driven decision-making.

Business Problem

The organization lacked a unified framework for understanding how production performance impacts financial outcomes. While production volumes were available, management required deeper insights into:

- Which fields and wells drive overall production?
- How operational factors affect BOE performance?
- How production performance translates into revenue and profit?
- What operational risks could threaten future profitability?

To address these challenges, an integrated Production and Financial Analytics Dashboard was developed.

Key Findings

1. Strong Production Performance Drives Financial Success

The portfolio produced approximately **30 million BOE**, generating over **\$1.26 billion in revenue** and approximately **\$1.15 billion in profit**.

This demonstrates a strong relationship between production output and financial performance, confirming that production remains the primary driver of revenue generation across the portfolio.

Business Impact

Higher BOE production directly contributes to higher revenue and profitability.

2. Production Is Concentrated in a Few Strategic Assets

Analysis of field and well performance revealed that **Agbami and Bonga** contribute the largest share of total BOE production.

These same fields also account for the highest revenue contribution.

Business Impact

The company's production and revenue streams are highly dependent on a small number of strategic assets.

Risk

Operational disruption within these assets could significantly affect overall company performance.

3. High-Producing Wells Are the Primary Value Drivers

A small group of wells contributes a substantial percentage of total BOE production.

These wells are responsible for generating the majority of revenue and profit across the portfolio.

Business Impact

Protecting these critical wells through proactive maintenance and operational monitoring is essential to sustaining future profitability.

4. Water Cut Is a Key Driver of Operational Performance

Water Cut analysis was included to monitor reservoir health and production efficiency.

The dashboard indicates that Water Cut remains an important operational metric because increasing water production can lead to:

- Higher treatment costs
- Increased operating expenses
- Reduced production efficiency

Business Impact

Water Cut directly influences operating costs and can negatively affect profitability if not properly managed.

5. Production Trends Influence Future Revenue Outlook

The BOE trend analysis shows changes in production levels over time.

Because revenue generation depends largely on production output, any sustained decline in BOE production could reduce future revenue and profit.

Business Impact

Monitoring production trends enables proactive intervention before financial performance is impacted.

6. Offshore Assets Generate Greater Financial Value

Revenue by region indicates that Offshore Deepwater operations contribute significantly more revenue than Onshore assets.

This reflects stronger production performance and greater asset value within offshore operations.

Business Impact

Offshore assets remain the primary revenue-generating segment of the portfolio.

Strategic Recommendations

Based on the analysis, the following recommendations are proposed:

- **Production Optimization**

Continue monitoring BOE trends to identify early signs of production decline and support proactive intervention planning.

- **Asset Protection**

Prioritize maintenance and operational reliability programs for top-performing fields and wells, particularly Agbami and Bonga.

- **Water Management**

Implement targeted water-management strategies to reduce water handling costs and improve production efficiency.

- **Profitability Enhancement**

Focus capital investment on high-producing and high-revenue assets while continuously optimizing operating expenses.

- **Predictive Decision-Making**

Leverage production and reservoir performance indicators to forecast future asset performance and improve operational planning.

Conclusion

The analysis demonstrates a strong connection between operational performance and financial outcomes. Fields and wells that generate the highest BOE production also contribute the greatest share of revenue and profitability. While the portfolio remains financially strong, continued focus on production optimization, asset reliability, and water management will be critical to sustaining long-term value creation and maximizing asset profitability.